

JKBOSE 2026

CLASS 11TH

#XUV 

REDOX REACTIONS

LECTURE-03 CHEMISTRY

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ON
TOP



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Topics To Covered

- 1 Calculation of Oxidation Number**
- 2 Types of Redox Reactions**
- 3 Balancing The Redox Reactions**

Oxidation No. $\rightarrow$

- 1) Free State / Elementary state $\rightarrow 0$
- 2) Charge on ion = ON
- 4) Metal and non-metal bond. ($\overset{+}{\text{Na}}(\overset{-}{\text{Cl}})$)
- 5) Group 1 (+1) & Group 2 (+2).
- 6) H mostly +1, but in metal hydrides (-1)
- 7) O mostly -2, but in H_2O_2 , $\text{Na}_2\text{O}_2 = -1$ & RbO_2 , $\overset{\uparrow -1/2}{\text{CaO}_2}$. $\text{O}_2\text{F}_2 = +1$ & $\text{OF}_2 = +2$.
- 8) F only -1 other halogens may show others.
- 9)

$$\underline{\vec{O}F_2}^x$$

$$a) -2$$

$$b) -1$$

$$c) -\frac{1}{2}$$

$$d) +2$$

$$\vec{O}F_2^x$$

$$x + 2(-1) = 0$$

$$x - 2 = 0$$

$$x = \textcircled{+2}$$



$$1 + 3 + 2x + 8(-2) = 0$$

$$4 + 2x - 16 = 0$$

$$2x + 4 - 16 = 0$$

$$2x - 12 = 0$$

$$2x = 12$$

$$x = \frac{12}{2} = (+6)$$

$$\overset{x}{(\underline{SO}_4)^{2-}}$$

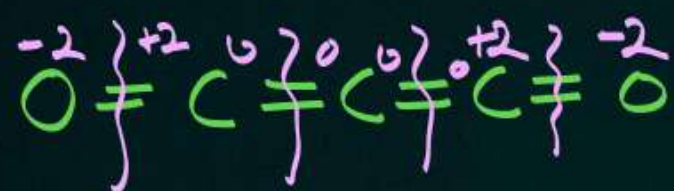
$$x + 4(-2) = -2$$

$$x - 8 = -2$$

$$x = 8 - 2$$

$$x = \underline{+6}$$

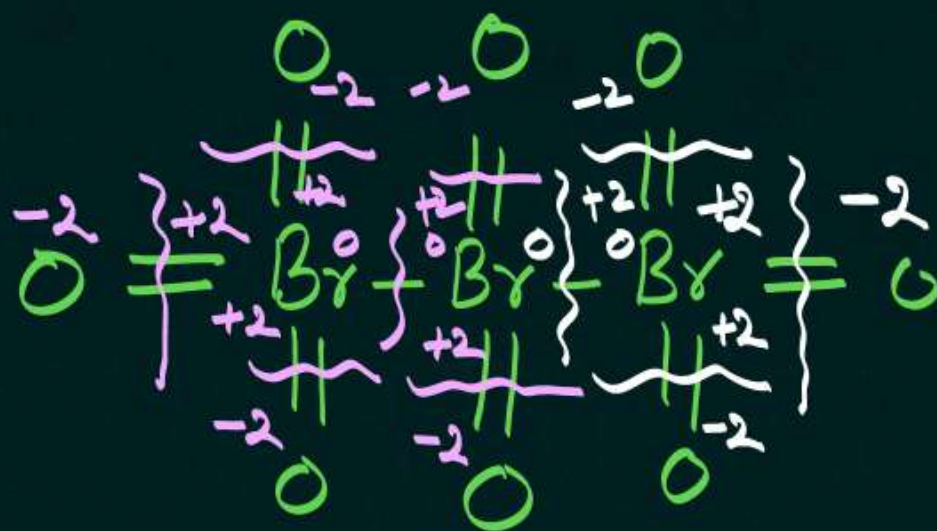
SPECIAL CASES.



Two O = -2.

Two C = +2

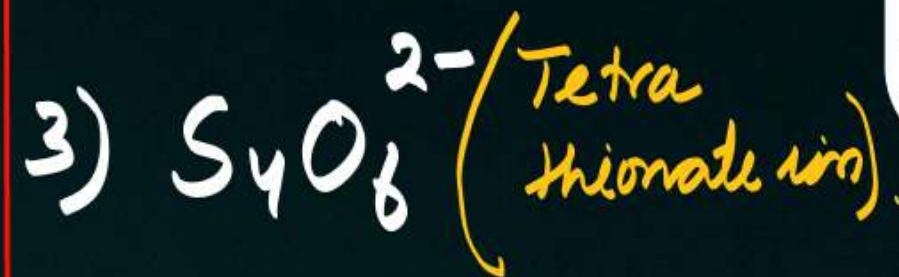
One C = 0.



Eight O = -2.

Two Br = +6.

One Br = +4.



Two S = +5

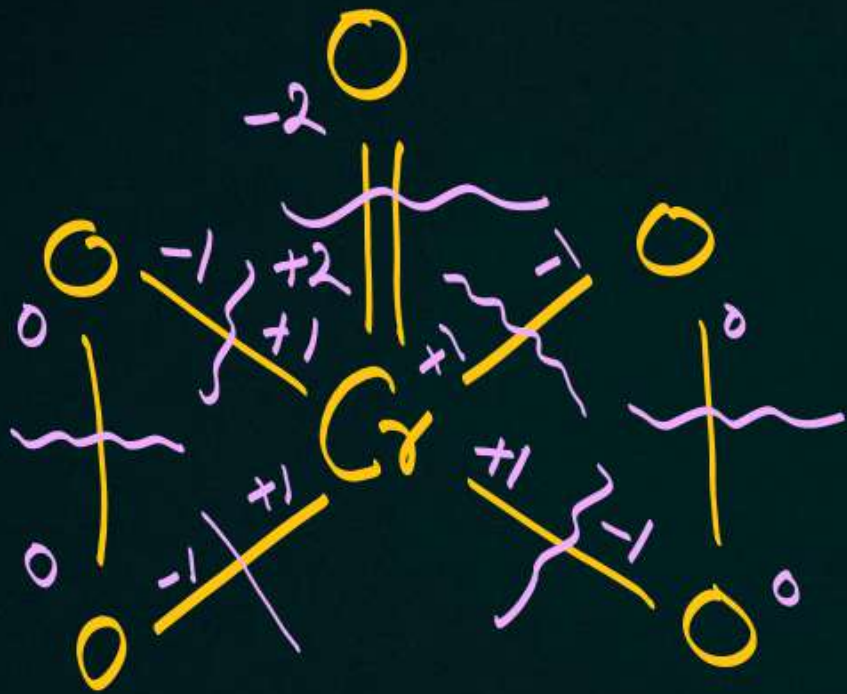
Two S = 0.

Six O = -2



SPECIAL CASES.

4) CrO_5



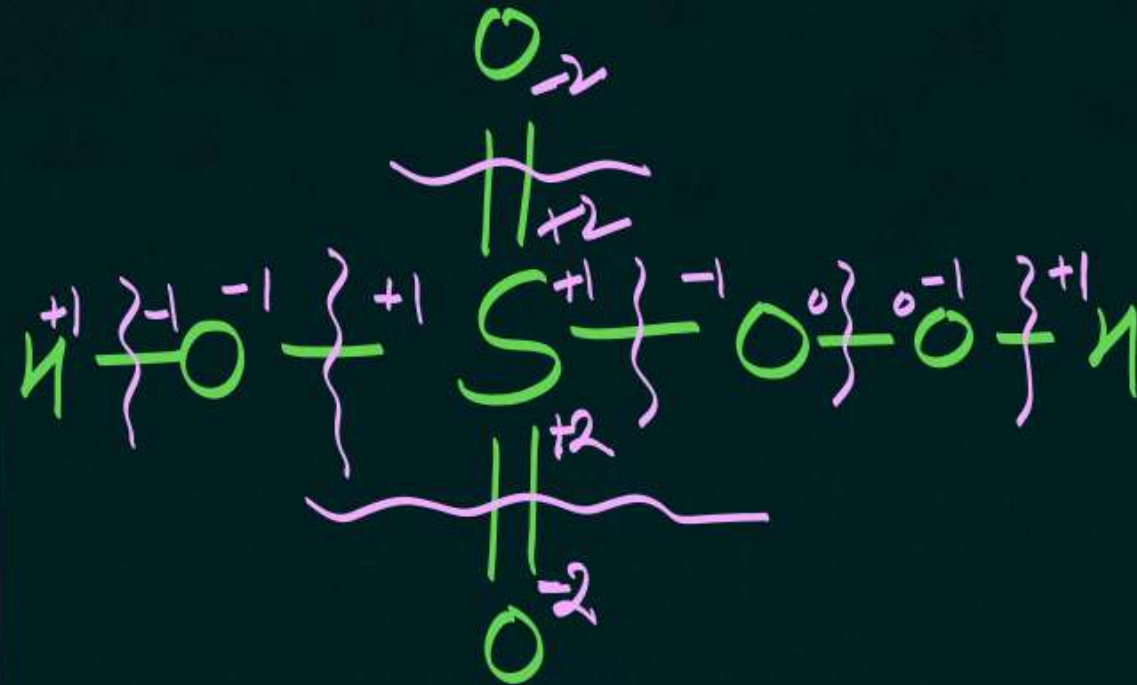
$\text{Cr} = +6$

One $\text{O} = -2$

Four $\text{O} = -1$



5) H_2SO_5 (Caros Acid)

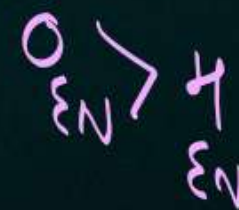


$\text{S} = +6$

Three $\text{O} = -2$

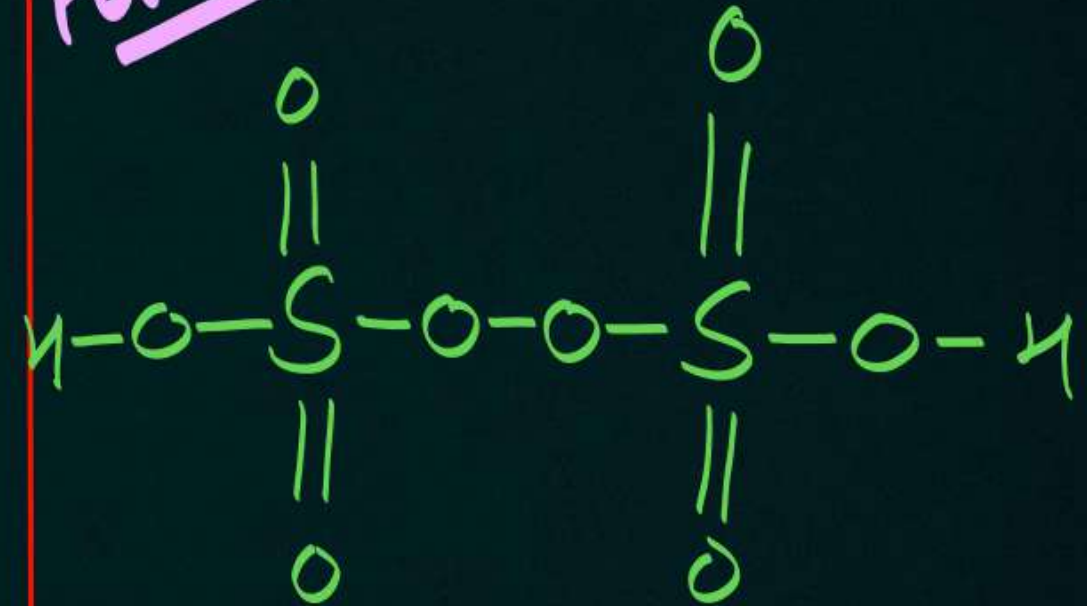
Two $\text{O} = -1$

Two $\text{H} = +1$



6) $\text{H}_2\text{S}_2\text{O}_8$

H.W For COMMUNITY (Marshall's Acid)



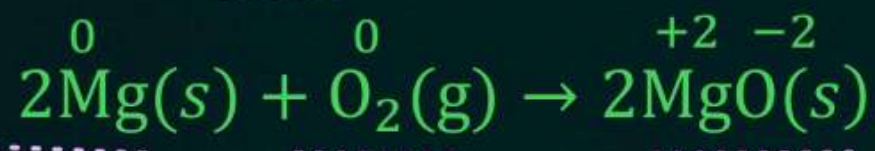
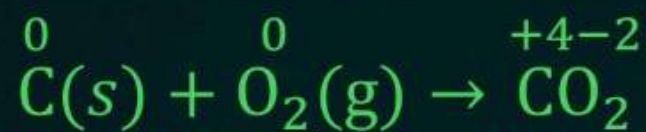
Types of Redox Reactions

1. Combination reactions

The chemical reactions in which two or more substances (elements or compounds) combine to form a single substance are called combination reactions. A combination reaction may be expressed as:



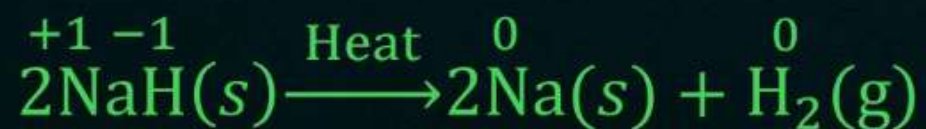
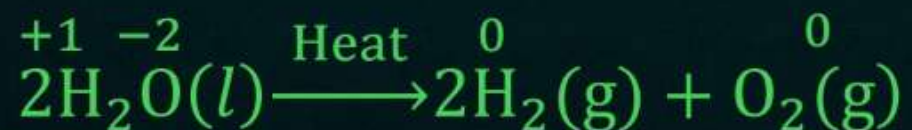
Some common combination redox reactions are:



2. Decomposition reactions

The chemical reactions in which a compound breaks up into two or more simpler substances are called decomposition reactions.

Some common examples of redox decomposition reactions are:



Types of Redox Reactions



3. Displacement reactions

less reactive element

more reactive element

The reactions in which one ion (or atom) in a compound is replaced by an ion (or atom) of other element are called displacement reactions.

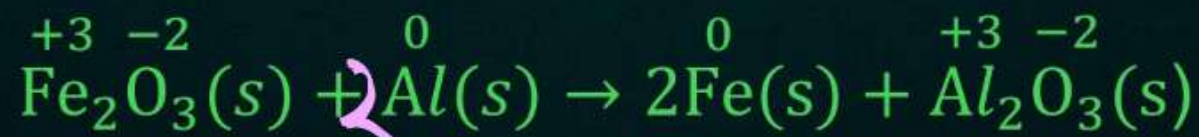
The displacement reactions are of two types:

- (a) Metal displacement and
- (b) Non-metal displacement

(a) Metal displacement reactions.

In these reactions, a metal in a compound can be displaced by another metal in the uncombined state.

Some common examples of metal displacement reactions are:



$\text{A} > \text{B}$
more reactive

Activity Series

OR

Reducing Agents

(Loss of e^-)



Oxidation

K
Na
Ca
Mg
Al
Zn
Fe
Pb
H
Cu
Hg
Ag
Au
Pt

↓
decrease

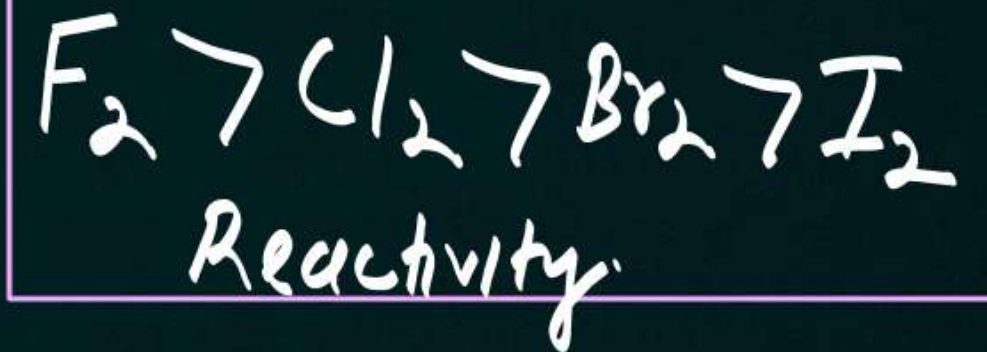
→ e^- donate (Oxidation) → Reducing Agent

Reactivity Series

- Reactivity **decreases** from top to bottom.
- Elements at the top are **more reactive**.
- They lose electrons more easily, so they are **strong reducing agents**.
- As we move down the series, the reactivity and reducing power **decrease**.
- Elements at the bottom are least reactive and are **weak reducing agents**.



A diagram illustrating the formation of non-metal compounds. On the left, the words "Non-metal" and "Metal" are written. Two arrows originate from these words and point towards the right, where the words "Non-Metal compound" are written and enclosed in an oval.

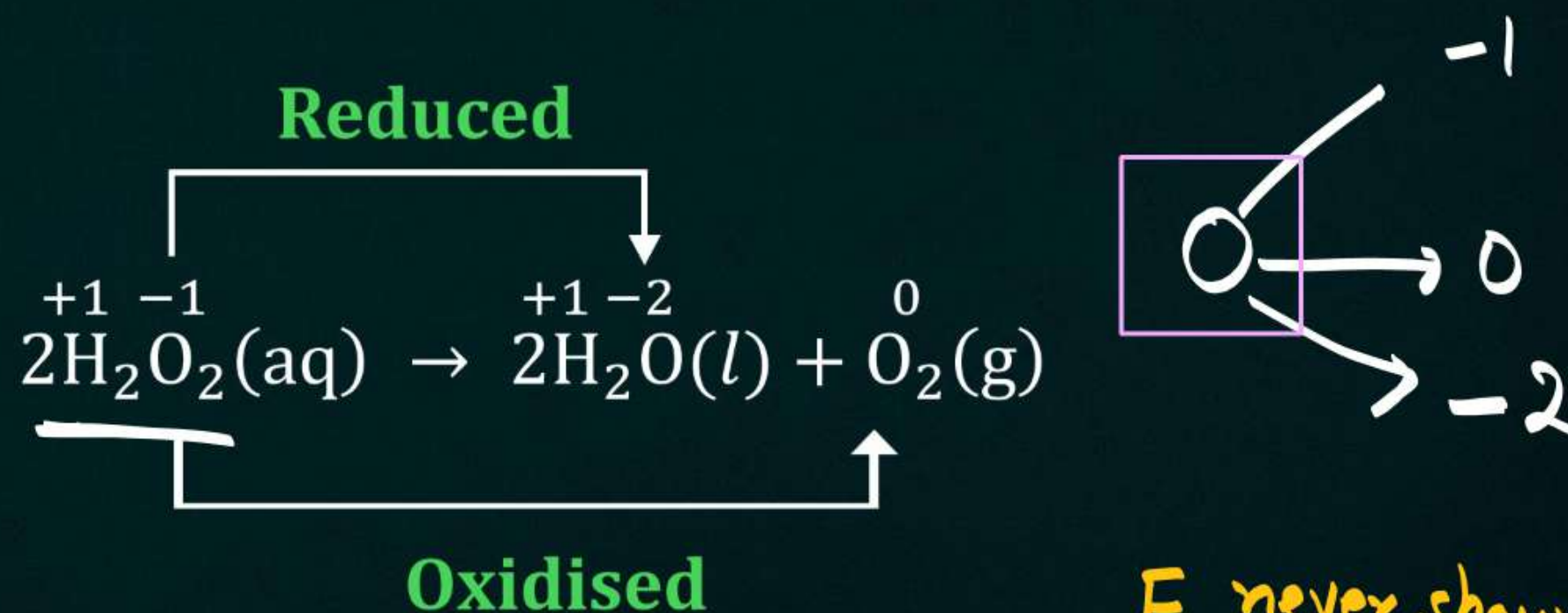
$$\begin{array}{l} \overset{0}{2\text{Na}}(\text{s}) + \overset{+1-2}{2\text{H}_2\text{O}}(\text{l}) \rightarrow \overset{+1-2+1}{2\text{NaOH}}(\text{aq}) + \overset{0}{\text{H}_2}(\text{g}) \\ \underline{\text{M}} \qquad \qquad \underline{\text{NM}} \\ \overset{0}{\text{Ca}}(\text{s}) + \overset{+1-2}{2\text{H}_2\text{O}}(\text{l}) \rightarrow \overset{+2-2+1}{\text{Ca}(\text{OH})_2}(\text{aq}) + \overset{+0}{\text{H}_2}(\text{g}) \\ \underline{\text{M}} \qquad \qquad \underline{\text{NM}} \end{array}$$


Types of Redox Reactions



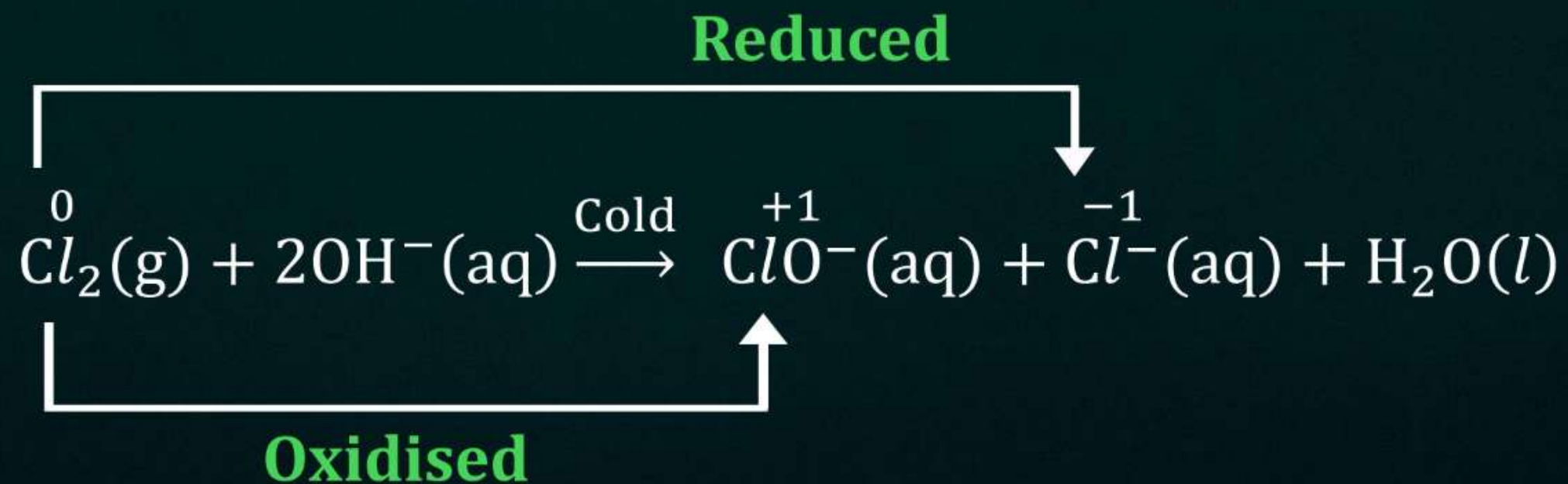
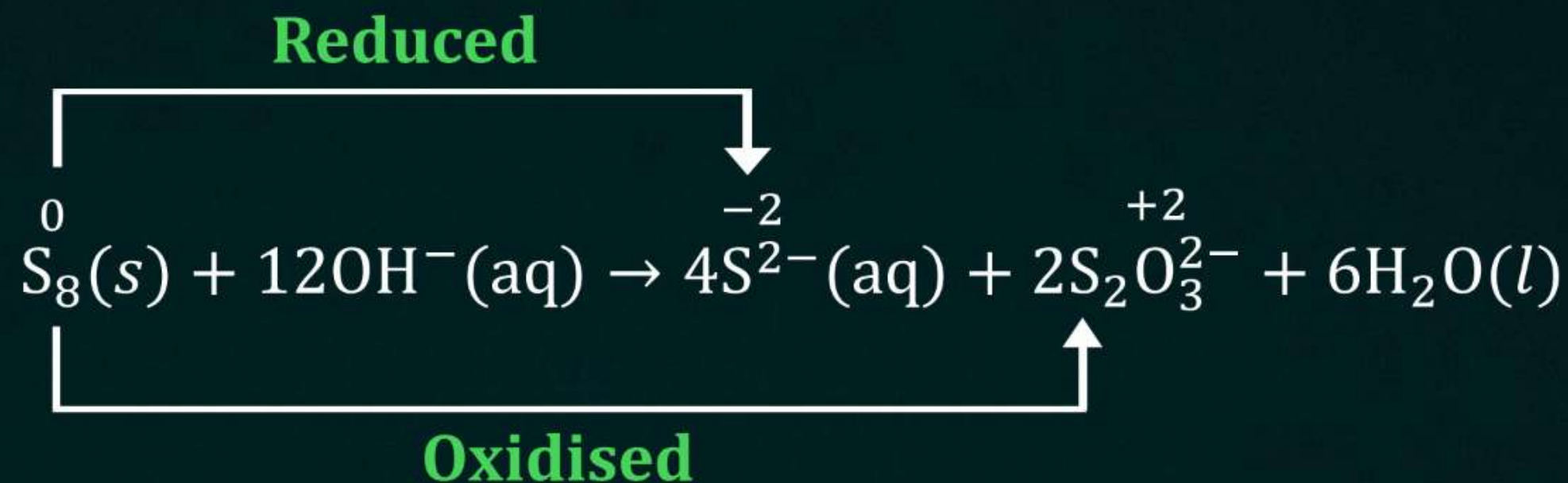
4. Disproportionation reactions

A reaction in which the same species is simultaneously oxidised as well as reduced is called a disproportionation reaction. For such redox reactions to occur, the reacting species must contain an element which has at least **three** oxidation states.



F never show
Disproportion rxn.
Because O.N of F = -1, 0

Types of Redox Reactions



HOW TO DETERMINE MAXIMUM AND MINIMUM OXIDATION STATE →

- Maximum O.S = No. of valence e^- .
- Minimum O.S = No. of valence e^- - 8.

e.g. In case of sulphur.

$$\text{Maximum O.S} = \text{No. of valence } e^- = (+6)$$

$$\text{Minimum O.S} = 6 - 8$$

$$= (-2)$$



NOTE :→

- If the element is in its max. O.S = behave as Oxidising Agent.
- If the element is in least Oxidation state, behave as Reducing Agent.
- If the element is in O.S which is in between Max & Minimum, then it behave as Oxidising as well as Reducing agent.

Q. Calculate Max & minimum O.N of P.

Sol. Max. O.S = No. of Valence $\bar{e}$ = $\boxed{+5}$

Min. O.S = Valence $\bar{e}$ - 8

$$= 5 - 8 = \boxed{-3}$$

$\boxed{15}$
N
P

Q. Calculate the Max & Min. O.S of Cl ^{17 (Halogen)}

a) +6, -2.

Hint: →

Max O.S = No. of Valence e^- = +7

Min O.S = Valence - 8 = 7 - 8 = **-1**

b) +7, -1

c) +7, -2

d) +8, +3.



Question



S $\rightarrow$ Max O.S. = +6
Min O.S. = -2
 $\textcircled{+4}$

$\overset{2+}{\text{H}_2}\overset{-2}{\text{S}}$ acts only as reductant where as SO_2 acts as both oxidant and reductant. Why? (PYQ)

Answer:

1) H_2S acts only as reductant:

Oxidation state of S in $\text{H}_2\text{S} = -2$

- This is the minimum oxidation state (lowest possible) for sulfur.
- So, H_2S can only lose electrons (get oxidized) and cannot gain electrons any further.
- Therefore, H_2S acts only as a Reductant.

2) SO_2 acts as both oxidant and reductant:

Oxidation state of S in $\text{SO}_2 = +4$

- This is an intermediate oxidation state (between minimum -2 and maximum +6).
- SO_2 can either gain electrons (get reduced) to lower its oxidation state (to -2), so it acts as an Oxidising Agent.
- Or, SO_2 can lose electrons (get oxidized) to increase its oxidation state (to +6), so it acts as a Reducing Agent.

Balancing The Redox Reactions

(A) Oxidation Number Method

The various steps involved in the balancing of redox equations by oxidation number method are:

Step 1. Write the skeletal equation of all the reactants and products of the reaction.

Step 2. Indicate the oxidation number of each element above its symbol and identify the elements which undergo a change in the oxidation number (O.N.).

Step 3. Calculate the increase or decrease in oxidation number per atom and equalise it by multiply with suitable number.

Step 4. Balance all atoms other than H and O.

X **Step 5.** Finally balance H and O atoms by adding H_2O molecules using hit and trial method.

Balancing The Redox Reactions

Step 6. In case of ionic reactions,

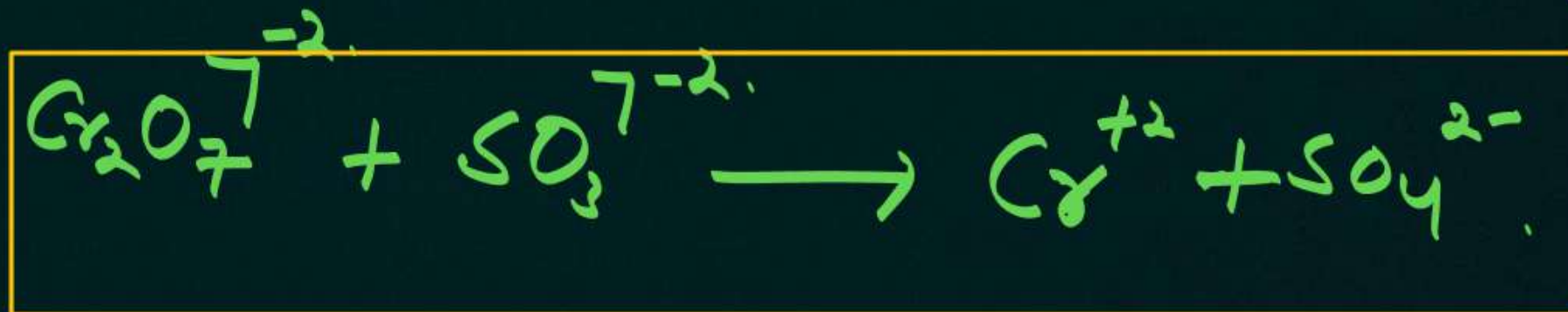
- (a) **For acidic medium.** First balance O atoms by adding H_2O molecule to whatever side deficient in O atoms and then balance H atoms by adding H^+ ions to whatever side deficient in H atoms.
- (b) **For basic medium.** First balance O atoms by adding H_2O molecules to whatever side deficient in O atoms. The H atoms are then balance by adding H_2O molecules equal in number to the deficiency of H atoms and an equal number of OH^- ions are added to the opposite side of the equation.

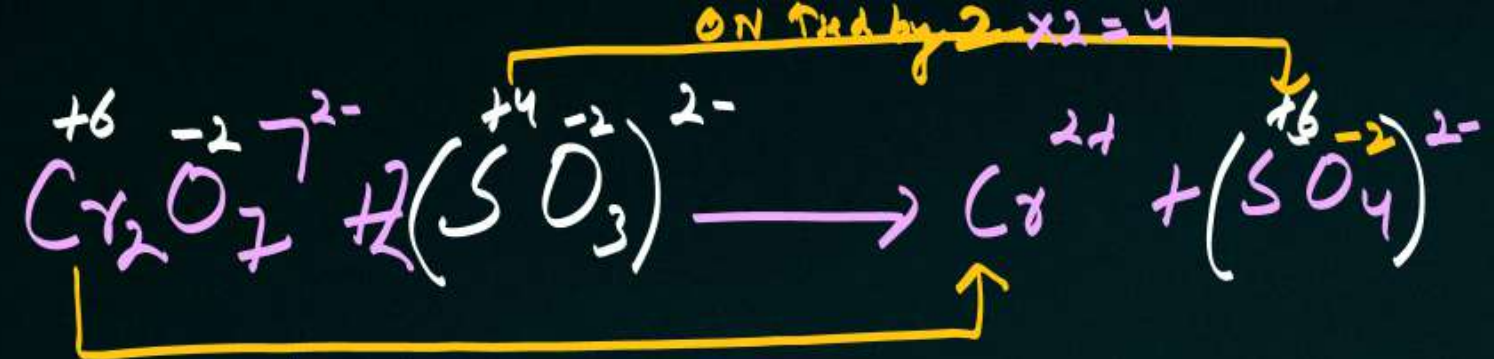
QUESTION



Write the balanced ionic equation for the reaction of potassium dichromate (VI), $K_2Cr_2O_7$, with sodium sulphite, Na_2SO_3 in **acid medium** to give chromium(II) ion and sulphate ion.

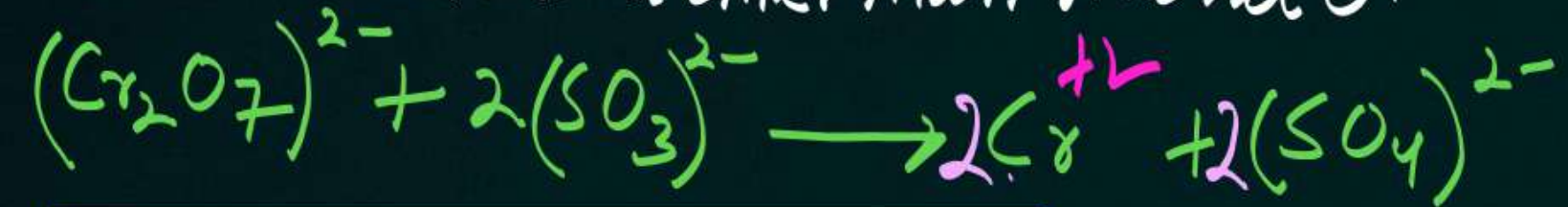
Sol. Skeletal Equation: $\overset{(2+)(-2.)}{\underline{K_2Cr_2O_7}} + \overset{+2(-2)}{Na_2SO_3} \longrightarrow Cr^{+2} + SO_4^{2-}$





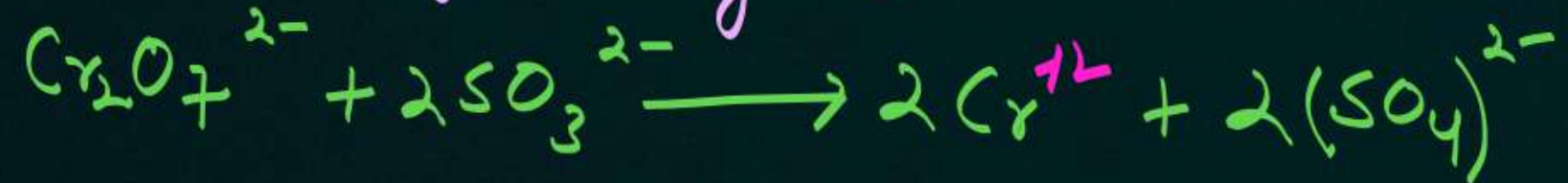
ON. led by 4

- Balance all atoms other than H and O.

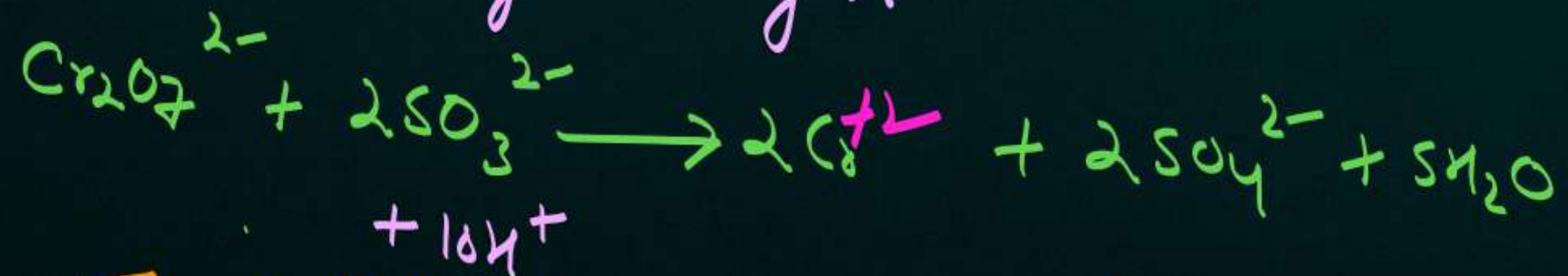


In Acidic Medium

- Balance 'O' by adding H_2O .



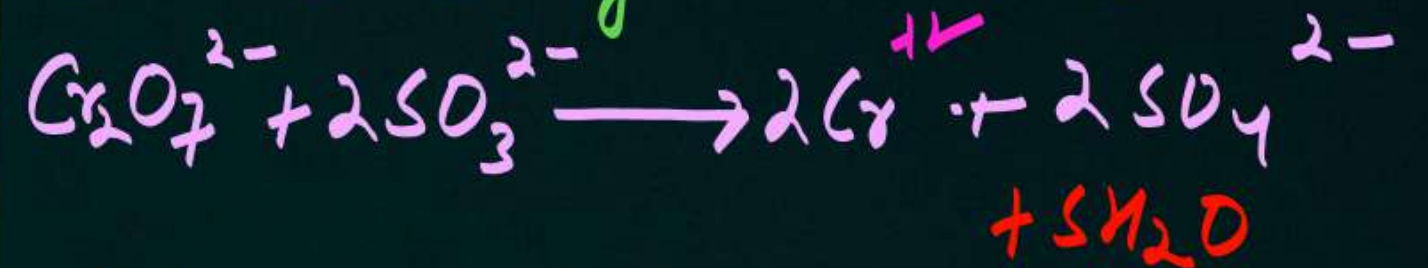
- Balance 'H' by adding H^+



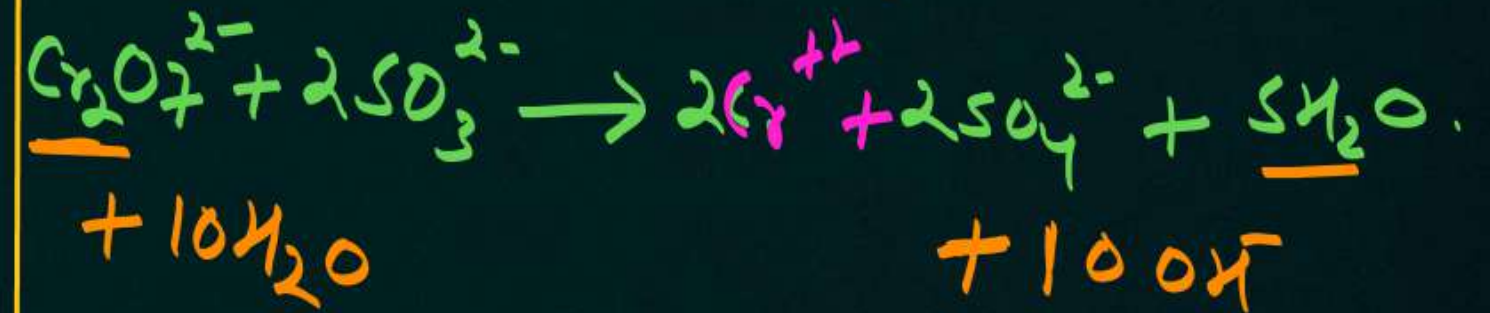
Thus is Balanced.

Basic Medium

- Balance 'O' by adding H_2O .



- Balance the 'H' by adding H_2O and add equal no. of OH^- to opposite side.

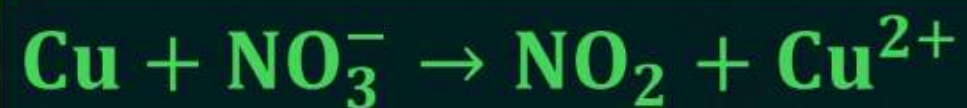


Thus is Balanced.

QUESTION



Copper reacts with nitric acid. A brown gas is formed and the solution turns blue.
The equation may be written as:

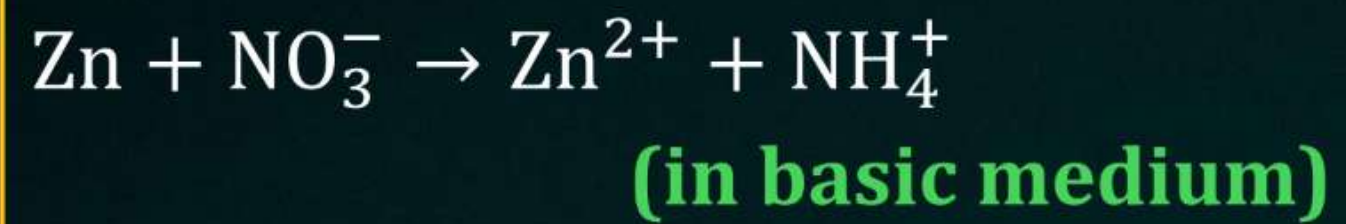


Balance the equation by oxidation number method.

Audier

Basie

QUESTION



QUESTION



The oxidation state of Cr in CrO_5 is:

[PYQ]

1 +10

2 -10

3 +6

4 -6

QUESTION

The oxidation number of the two Cl atoms in bleaching powder, $CaOCl_2$ are: **[PYQ]**

- 1 $-1, -1$
- 2 $+1, -1$
- 3 $+1, +1$
- 4 $0, -1$

QUESTION



Which of the following elements does not show disproportionation tendency?

1 *Cl*

2 *Br*

3 *F*

4 *I*

QUESTION

Identify disproportionation reaction.



- ① Mohd. Salik
Munshi
- ② Mohd. Abbass
- ③ Numaan Wani

We will do it together.

Thank

You

#Teelenge.